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Division: A2

ASSIGNMENT No: 9

TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

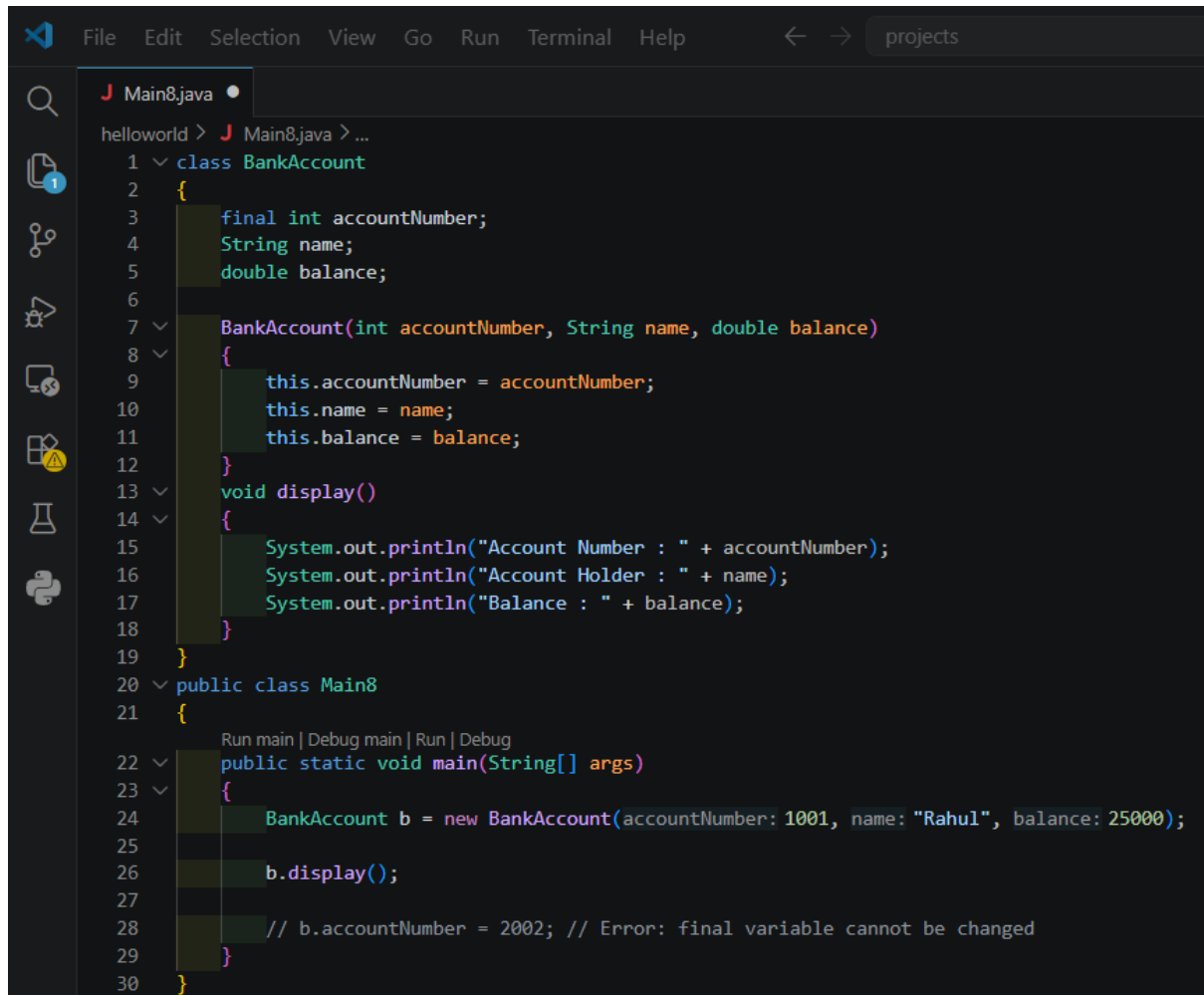
Learning Objectives: To implement the final keyword with variables, methods, and classes.

Theory : The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Exercise

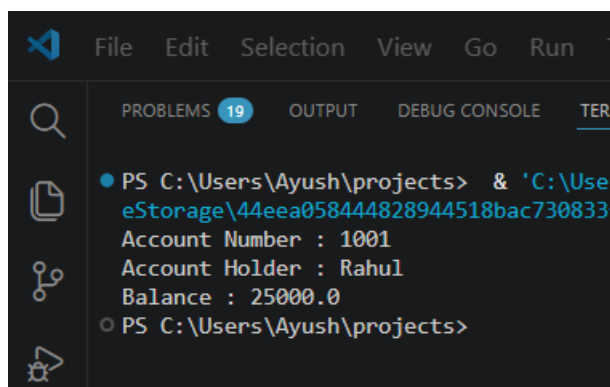
1. Create a Bank Account program where account number is final and cannot be changed.

Program:



```
File Edit Selection View Go Run Terminal Help
Main8.java
helloworld > J Main8.java > ...
1 class BankAccount
2 {
3     final int accountNumber;
4     String name;
5     double balance;
6
7     BankAccount(int accountNumber, String name, double balance)
8     {
9         this.accountNumber = accountNumber;
10        this.name = name;
11        this.balance = balance;
12    }
13    void display()
14    {
15        System.out.println("Account Number : " + accountNumber);
16        System.out.println("Account Holder : " + name);
17        System.out.println("Balance : " + balance);
18    }
19 }
20 public class Main8
21 {
22     Run main | Debug main | Run | Debug
23     public static void main(String[] args)
24     {
25         BankAccount b = new BankAccount(accountNumber: 1001, name: "Rahul", balance: 25000);
26         b.display();
27
28         // b.accountNumber = 2002; // Error: final variable cannot be changed
29     }
30 }
```

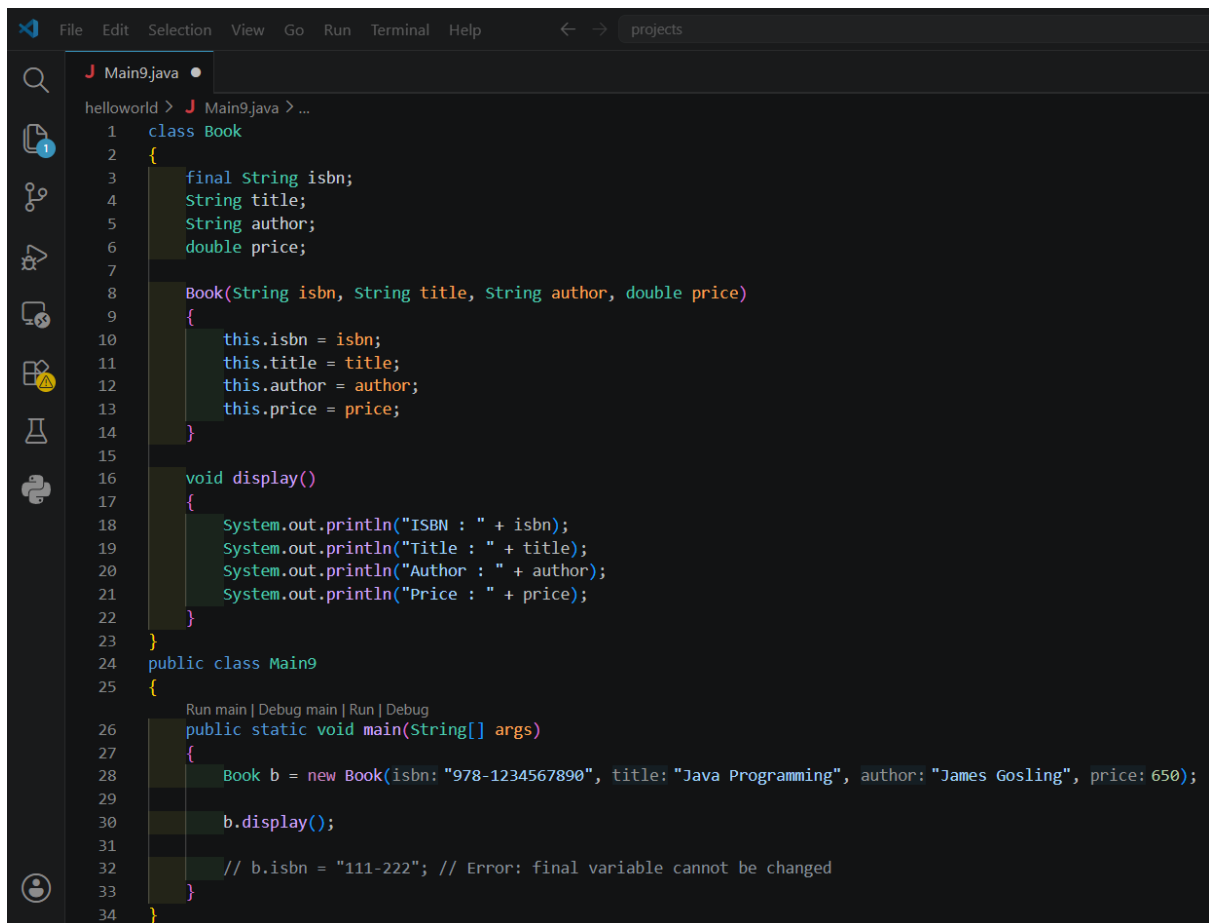
Output:



```
File Edit Selection View Go Run
PROBLEMS 19 OUTPUT DEBUG CONSOLE
PS C:\Users\Ayush\projects> & 'C:\Use
eStorage\44eea058444828944518bac730833
Account Number : 1001
Account Holder : Rahul
Balance : 25000.0
PS C:\Users\Ayush\projects>
```

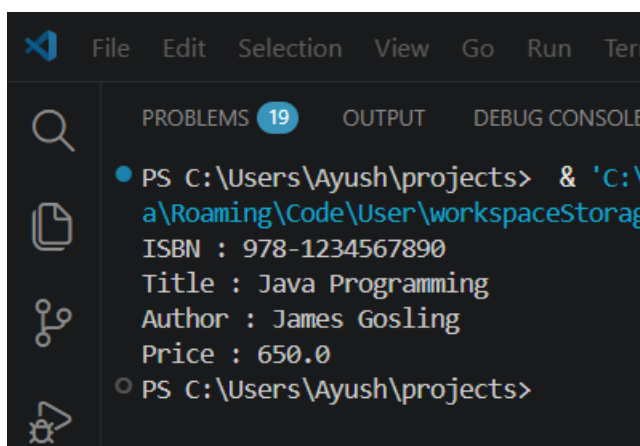
2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Program:



```
1 class Book
2 {
3     final String isbn;
4     String title;
5     String author;
6     double price;
7
8     Book(String isbn, String title, String author, double price)
9     {
10         this.isbn = isbn;
11         this.title = title;
12         this.author = author;
13         this.price = price;
14     }
15
16     void display()
17     {
18         System.out.println("ISBN : " + isbn);
19         System.out.println("Title : " + title);
20         System.out.println("Author : " + author);
21         System.out.println("Price : " + price);
22     }
23 }
24 public class Main9
25 {
26     Run main | Debug main | Run | Debug
27     public static void main(String[] args)
28     {
29         Book b = new Book(isbn: "978-1234567890", title: "Java Programming", author: "James Gosling", price: 650);
30
31         b.display();
32
33         // b.isbn = "111-222"; // Error: final variable cannot be changed
34     }
35 }
```

Output:



```
PS C:\Users\Ayush\projects> & 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' -cp 'C:\Users\Ayush\projects\Main9.class' Main9
ISBN : 978-1234567890
Title : Java Programming
Author : James Gosling
Price : 650.0
PS C:\Users\Ayush\projects>
```